

Telecom Customer Churn Analysis

Complete Project Documentation — From Overview to Dashboard

Telco Customer Churn Dataset — IBM/Kaggle sample

1. Project Overview

This project analyzes customer churn in a telecommunications company. The objective is to understand how customer characteristics, contract type, tenure, payment method, internet service, subscribed services, and monthly charges are associated with churn, then translate the findings into business priorities, retention recommendations, and an executive dashboard.

Project Goal

Identify and prioritize customers and segments with high churn risk in order to support targeted retention actions and reduce customer and revenue loss.

Business Problem

Customer churn reduces the customer base and recurring revenue. The analysis therefore focuses on where churn is concentrated, which segments deserve attention, and where revenue exposure is highest.

Project Workflow

1. Business Understanding
2. Data Acquisition & Documentation
3. Data Understanding
4. Data Quality Assessment
5. Data Cleaning & Preparation
6. Exploratory Data Analysis (EDA)
7. Required Business Questions
8. Our Own Questions
9. Diagnostic & Segmentation Analysis
10. Business Impact & Prioritization
11. Insight Validation
12. Recommendations
13. Dashboard
14. Final Presentation / Final QA

2. Stage 01 — Business Understanding

Item	Definition
Business domain	Telecommunications
Primary problem	Customer churn and retention
Target variable	Churn
Analytical objective	Understand churn patterns and prioritize retention opportunities
Business outcome	Reduce customer loss and protect recurring monthly revenue

3. Stage 02 — Data Acquisition & Documentation

The project uses the Telco Customer Churn dataset. The raw CSV is preserved as the reference source, while the cleaned CSV is used for analysis.

Raw path: D:\Data Analytics\Project\2) Dataset\1) Raw\WA_Fn-UseC_-Telco-Customer-Churn.csv

Cleaned path: D:\Data Analytics\Project\2) Dataset\2) Cleaned\telco_churn_cleaned.csv

The cleaned dataset contains 7,043 customer records and 26 source/derived fields used by the project:

customerid, gender, seniorcitizen, partner, dependents, tenure, phoneservice, multiplelines, internetservice, onlinesecurity, onlinebackup, deviceprotection, techsupport, streamingtv, streamingmovies, contract, paperlessbilling, paymentmethod, monthlycharges, totalcharges, churn, senior_citizen_label, churn_flag, tenure_group, monthly_charge_band, high_monthly_charge

4. Stage 03 — Data Understanding

This stage was intentionally diagnostic. The raw dataset was not modified.

- Loaded the raw CSV with Pandas.
- Inspected the first records and dataset shape.
- Reviewed exact column names.
- Inspected data types and non-null counts with info().
- Counted unique values.
- Explicitly inspected Contract, InternetService, PaymentMethod, and Churn categories.
- Calculated churn counts and percentages.
- Reviewed major categorical distributions.
- Calculated early descriptive churn rates.
- Reviewed numerical summaries with describe().
- Examined tenure, monthly charges, and TotalCharges distributions.
- Created preliminary boxplots to flag unusual observations for later quality assessment.

Observations at this stage were treated as descriptive signals, not causal conclusions.

5. Stage 04 — Data Quality Assessment

- Checked missing and blank-like values.
- Investigated TotalCharges because it can contain blank strings while being conceptually numeric.
- Checked categorical values against the actual file.
- Reviewed duplicate and structural consistency issues.
- Reviewed numerical distributions and potential outliers.
- Separated genuine unusual observations from actual data errors.

6. Stage 05 — Data Cleaning & Preparation

- Converted numeric fields to appropriate numeric types where required.
- Handled the TotalCharges numeric issue appropriately.
- Standardized field naming for analysis.
- Created churn_flag: Yes = 1 and No = 0.

- Created senior_citizen_label for readable demographic analysis.
- Created tenure_group for lifecycle segmentation.
- Created monthly_charge_band for customer-value segmentation.
- Created high_monthly_charge for high-charge segmentation.
- Verified the cleaned dataset after preparation.

7. Exploratory Data Analysis (EDA)

- Customer and feature counts.
- Average tenure and charges.
- Contract and churn distributions.
- Churn rate versus churn count.
- Tenure distribution.
- Monthly and total charge distributions.
- Initial comparisons of churned versus retained customers.
- Basic categorical and numerical visualizations.

8. Stage 07 — Required Business Questions

15. What is the churn rate by contract type?
16. What is the churn rate by payment method?
17. What is the churn rate by Internet Service?
18. How do tenure and monthly charges differ between churned and retained customers?
19. Which customers combine month-to-month contracts with high monthly charges?
20. Which subscribed services are most associated with churn?
21. At which tenure stage is churn highest?
22. Does contract type change the relationship between tenure and churn?
23. Which Contract × Payment Method combinations are highest risk?
24. Does the number of subscribed services relate to churn?
25. Which demographic groups have higher churn?
26. Who are the highest-risk customer segments?
27. Which high-value customers have churned?
28. What monthly revenue is represented by churned customers?
29. Which high-risk segments have the greatest revenue exposure?
30. What is the final executive ranking of priority segments?

9. Stage 08 — Our Own Questions

31. Does churn risk change significantly across the customer lifecycle?
32. Does the impact of monthly charges depend on contract type?
33. Which customers represent the greatest revenue risk?
34. Are high-value customers actually more loyal?
35. Does having more services make customers more loyal?
36. Which service combinations are associated with the highest churn?
37. Which customer segment should the company prioritize first?
38. Can we create a practical Customer Risk Profile?

The exploratory Customer Risk Profile used observable signals such as month-to-month contract, short tenure, high monthly charges, no tech support, and no online security. It was treated as an analytical profile, not as a predictive machine-learning model.

10. Stage 09 — Diagnostic & Segmentation Analysis

10.1 Tenure Segmentation

Customers were bucketed into tenure groups and analyzed using customer count, churned customers, churn rate, and average monthly charge. A Plotly bar chart visualized churn rate by lifecycle stage.

10.2 Contract × Tenure

Churn rates were cross-tabulated by contract type and tenure group and visualized as a heatmap.

10.3 Contract × Payment Method

Churn rates were calculated for combinations of contract type and payment method. Segments with fewer than 50 customers were filtered for prioritization to avoid unstable results from very small groups.

10.4 Service-Level Segmentation

A service_count field was derived from Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, and Streaming Movies. Churn was then compared by number of subscribed services.

10.5 High-Risk Segments

Segments were evaluated using Contract × Tenure Group × Internet Service, with customer volume, churned customers, churn rate, and average monthly charge.

10.6 Revenue at Risk

Monthly revenue represented by churned customers was calculated as the sum of monthly charges for customers with churn_flag = 1.

10.7 Revenue Exposure by Segment

Estimated monthly revenue exposure was calculated as customers × average monthly charge. This supports prioritization by financial impact rather than churn rate alone.

10.8 Executive Risk Ranking

A ranking score combining churn rate and customer volume was used for analytical prioritization. The objective was to identify segments that are both materially large and associated with elevated churn.

11. Stage 10 — Business Impact & Prioritization

The project moved from descriptive findings to business prioritization. The highest churn percentage is not automatically the highest business priority.

- Consider churn rate.
- Consider number of customers affected.
- Consider average monthly charge.
- Consider estimated monthly revenue exposure.
- Use a minimum customer-volume threshold.
- Prioritize segments where churn risk and financial exposure overlap.

12. Stage 11 — Insight Validation

- Compared churn rates with customer counts.
- Checked findings across multiple categorical combinations.
- Compared churned and retained groups using averages and medians where appropriate.
- Applied customer-volume thresholds for segment prioritization.
- Distinguished association from causation.
- Validated revenue exposure calculations against the churned population.
- Used Python/Pandas as the primary analytical workflow and SQL as a secondary validation/query layer.

13. Role of SQL

SQL is used mainly as a secondary validation and business-query layer after the cleaned dataset is available in a database table. Python/Pandas remains the primary analysis workflow.

Example validation queries:

```
SELECT COUNT(*) AS Customers FROM TelcoChurn;
```

```
SELECT Contract, COUNT(*) AS Customers FROM TelcoChurn GROUP BY Contract;
```

14. Stage 12 — Recommendations

- Prioritize retention efforts toward month-to-month customers.
- Pay special attention to short-tenure customers during the early lifecycle.
- Review high monthly-charge customers when they also show other churn-associated characteristics.
- Investigate service and support gaps, especially absence of online security and technical support.
- Use contract × payment combinations for targeted retention campaigns.
- Prioritize using both churn risk and revenue exposure.
- Monitor churn and revenue-at-risk KPIs after retention actions are implemented.

15. Stage 13 — Dashboard

The dashboard is a separate worksheet from Raw Data, Cleaned Data, and Analysis. It is the presentation layer that communicates the most decision-relevant results.

Dashboard Goal

Identify and prioritize customers and segments with high churn risk in order to support targeted retention actions and reduce customer and revenue loss.

Final KPI Set

KPI	Why it matters	Calculation
Total Customers	Size of customer base	Count of CustomerID
Churned Customers	Customers lost	Count of CustomerID where Churn = Yes
Overall Churn Rate	Core churn measure	Average of churn_flag
Retained Customers	Customers retained	Count of CustomerID where Churn = No
Monthly Revenue at Risk	Monthly revenue represented by churned customers	Sum of MonthlyCharges where Churn = Yes

Average Monthly Charge	Average monthly customer value	Average of MonthlyCharges
------------------------	--------------------------------	---------------------------

High-Risk Customers was removed from the final KPI cards because the cleaned dataset does not contain an official risk_level field. The project is churn analysis rather than a predictive risk model. Risk is therefore better communicated through segmentation charts and tables.

Dashboard Questions / Visuals

- How does churn vary by contract type?
- At which tenure stage is churn highest?
- Which payment methods show higher churn?
- Which internet service groups show higher churn?
- Which Contract × Payment Method combinations are most risky?
- Which customer segments have the greatest revenue exposure?
- How does churn vary across monthly charge bands?
- How does churn vary by number of subscribed services?

Recommended Excel Workbook Structure

39. Dashboard — first/visible presentation sheet.
40. Raw_Data — original source, never modified.
41. Cleaned_Data — final prepared dataset.
42. Analysis — calculations and analytical outputs.
43. Dashboard_Calculations / Lookups — optional helper area for Pivot Tables and dashboard formulas.

16. KPI Calculation Using Excel Pivot Tables

KPI	Pivot setup	Calculation
Total Customers	Values: customerid	Count / Distinct Count
Churned Customers	Rows: churn; Values: customerid	Use Yes row
Retained Customers	Rows: churn; Values: customerid	Use No row
Overall Churn Rate	Values: churn_flag	Average; format as %
Monthly Revenue at Risk	Rows: churn; Values: monthlycharges	Sum; use Yes row
Average Monthly Charge	Values: monthlycharges	Average

17. Key Analytical Principles

- Raw data is preserved and never modified.
- Cleaning is separated from analysis.
- EDA is descriptive and does not establish causality.
- Segment rankings consider customer volume.
- Revenue exposure is considered alongside churn rate.
- The dashboard is a presentation layer, not a cleaning layer.
- Dashboard questions are selected from the broader analysis for decision relevance.
- High-Risk Customers is not a final KPI without a formal stored risk definition.
- Python/Pandas is the primary analysis tool; Plotly and Seaborn support visualization; Excel/Pivot Tables support dashboard presentation; SQL provides secondary validation.

18. Final Project Flow

**Business Goal → Data Acquisition → Data Understanding → Data Quality → Cleaning →
EDA → Business Questions → Own Questions → Diagnostic & Segmentation →
Business Impact → Validation → Recommendations → Dashboard**

This document summarizes the analytical methodology and outputs developed for the Telecom Customer Churn project.